

SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☒	Shut off cooling water
☒	Shut off fuel oil
☐	Shut off lubricating oil
☒	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D01-01	Exhaust valve stud, screwing-in torque	200	Nm
D01-02	Starting valve stud, screwing-in torque	145	Nm
D01-03	Fuel valve stud, screwing-in torque	80	Nm
D01-06	Cylinder cover, complete	1670	kg
D01-08	Cooling jacket	40	kg
D01-09	Exhaust valve stud	28	kg
D13-01	Hydraulic pressure, mounting	1500	
D13-02	Hydraulic pressure, dismantling	1400-1650	

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90151	46	Grinding tool for exhaust valve seat
P90151	58	Milling and grinding wheel for fuel valve seat
P90151	60	Milling and grinding wheel for starting valve seat
P90151	83	Grinding handle
P90151	130	Lifting chains
P90151	201	Handle
P90161		Cylinder cover, tools
P91351	10	Hydraulic pump, pneumatically operated
P91351	46	Hose with unions (1500 mm), complete
P91351	58	Hose with unions (3000 mm), complete
P91351	130	9-way distributor block, complete
P91359		Torque Spanners

1. Set blocking device **A** on the main starting valve in the BLOCKED position.

Engage turning gear **B**.

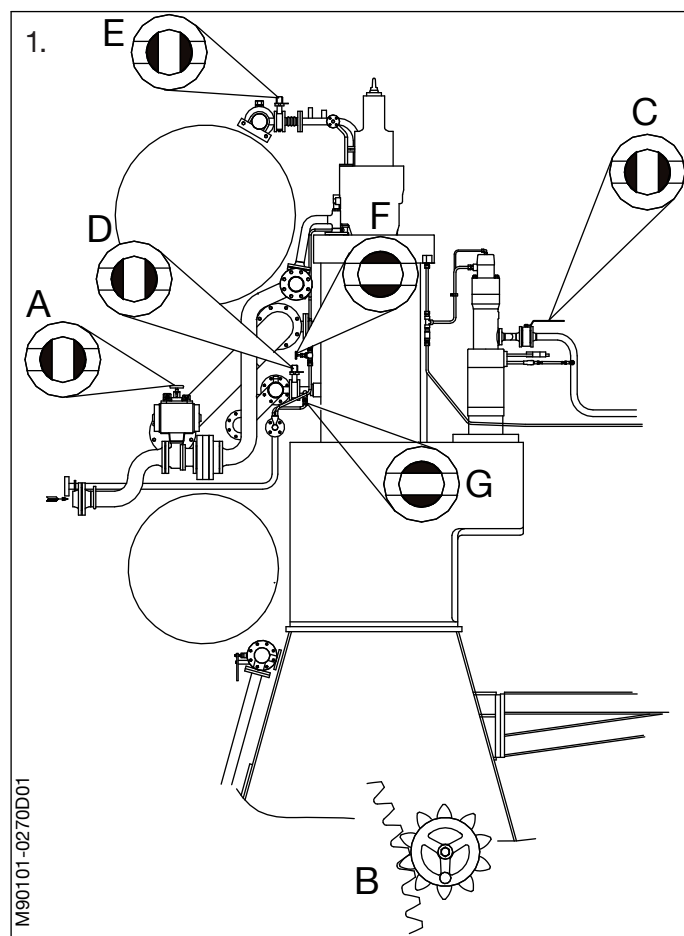
Close fuel oil inlet valve **C**.

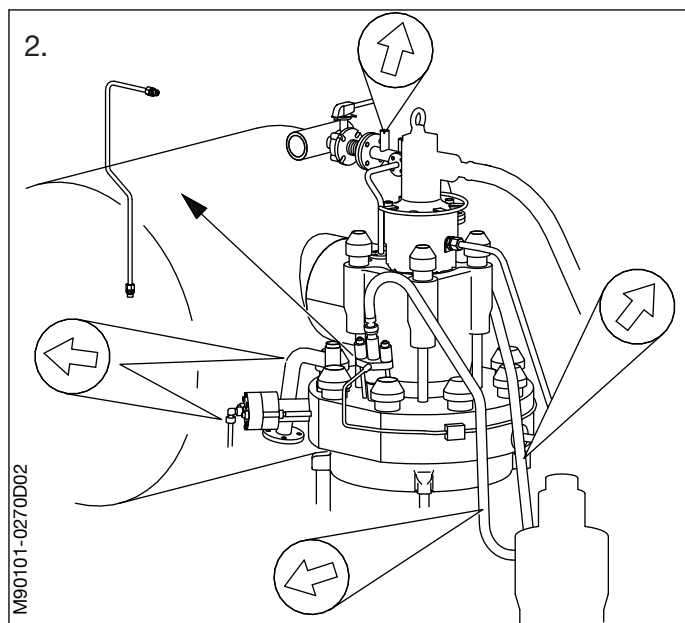
Close cooling water inlet valve **E**.

Open cooling water drain cocks **F** and **G**.

Shut off control air and safety air supply. Vent the manoeuvring system through the ball valve.

Also shut off the air supply on the pressure reduction unit for the hydraulic/pneumatic exhaust valve, and vent the system.





2. Remove the cooling water inlet and outlet pipes from the exhaust valve and cylinder cover.

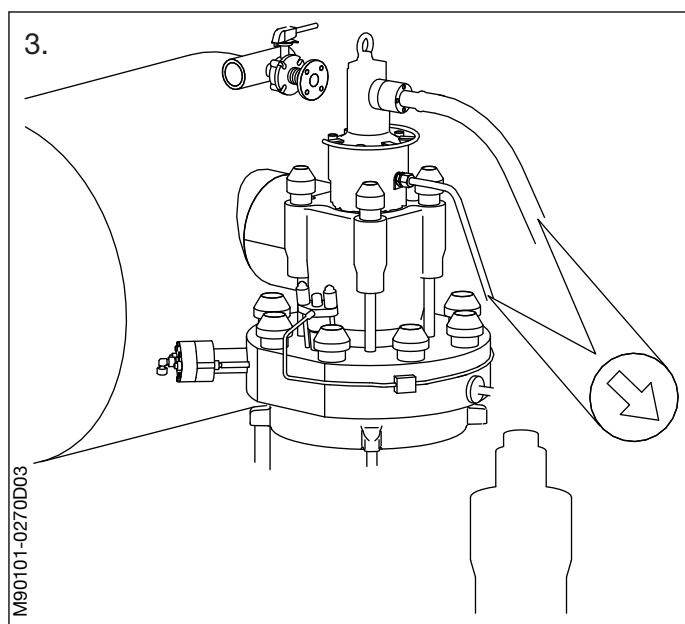
Remove the drain pipe from the intermediate pipe between the cylinder cover and the exhaust valve.

Dismount the fuel oil high-pressure pipes between fuel valves and fuel pump.

See Procedure 909-14.2.

Dismount the control air pipe from the starting air valve.

Remove the starting air pipe.



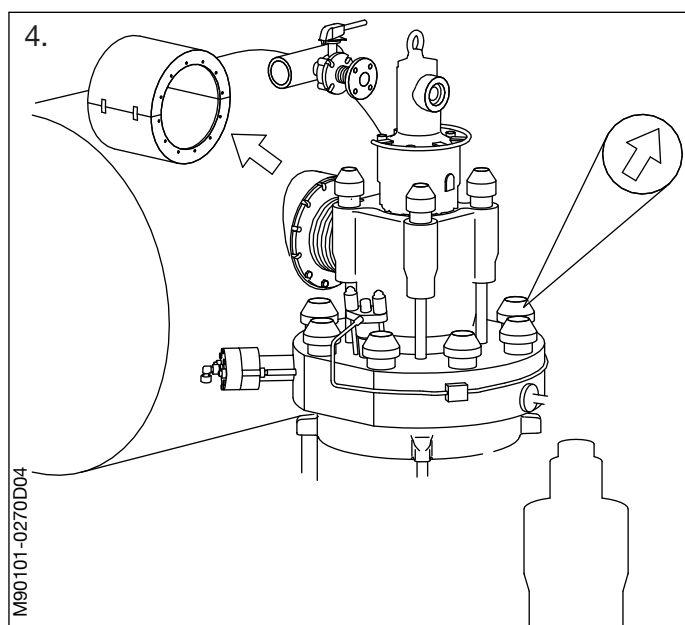
3. Remove the exhaust valve high-pressure pipe.

See Procedure 908-1.2.

Dismount the drain oil pipe between the exhaust valve and the hydraulic activator, the air pipe for pneumatic closing of the exhaust valve.

4. Remove the protective jacket enclosing the intermediate pipe between the exhaust receiver and the exhaust valve.

Dismount the screws in the flange between the intermediate pipe and the exhaust receiver. The four lowermost screws need not be dismantled, just loosened.



Remove the protective caps from the cylinder cover nuts.

5. Place eight hydraulic jack supports, one around each nut, and screw the eight hydraulic jacks onto the studs.

Connect the high-pressure pump to the jacks by means of the distributor block and the high-pressure hoses.

Using the vent screws of the hydraulic jacks, vent the system and increase the pressure as stated in Data.

For operation of the hydraulic tools, see Procedure 913-1.

Loosen the nuts by applying the tommy bar through the slots in the hydraulic jack supports.

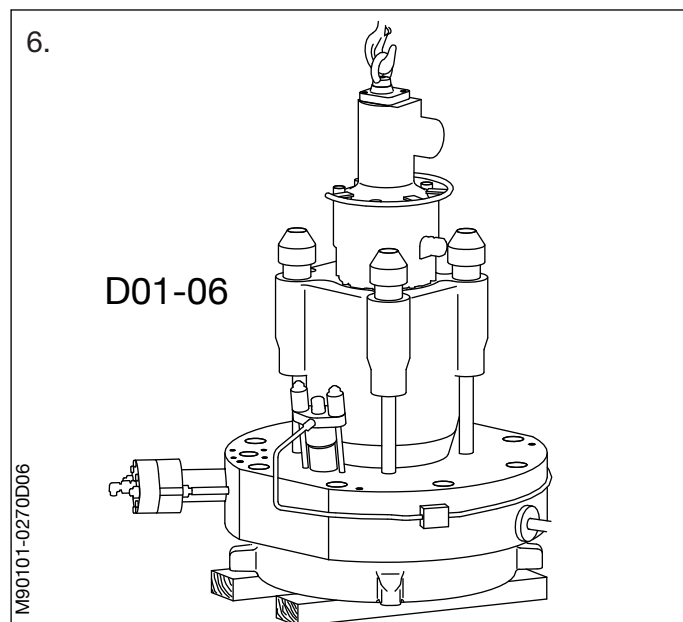
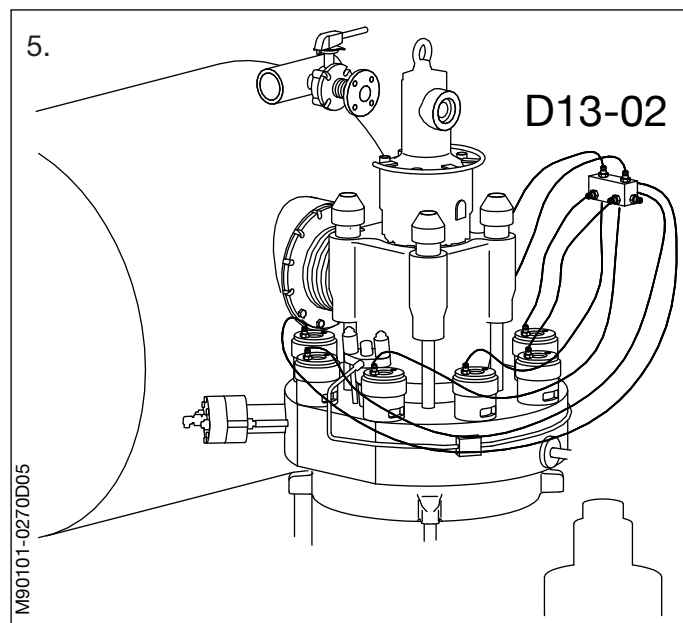
Relieve the pressure.

Disconnect the high-pressure hoses from the jacks.

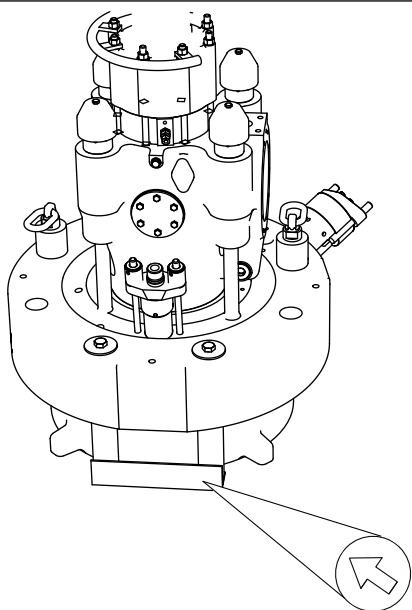
Remove the hydraulic jacks and their supports, and unscrew the nuts.

6. Hook the engine room crane on the lifting attachment on top of the exhaust valve.

Lift away the cylinder cover complete, and land it on a couple of wooden planks.



8.

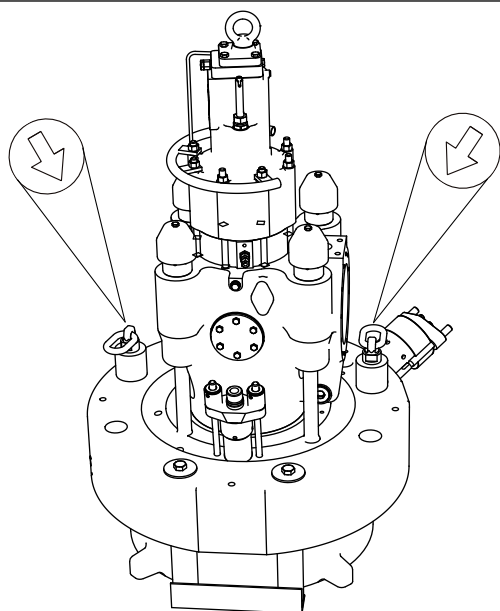


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When inspecting/cleaning the combustion chamber.

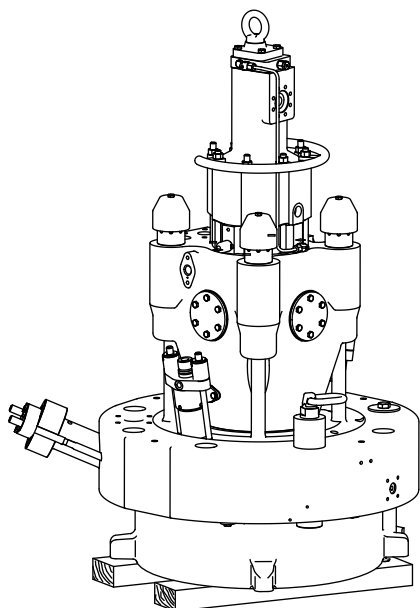
7. Lift the cylinder cover with the engine room crane.
8. Mount the tilting tool in the two holes for the cylinder cover studs close to the 'neighbouring' cylinder cover, opposite to the starting valve.
9. Mount the lifting tool in the two cylinder cover stud holes on the opposite side of the exhaust valve housing, as shown.
10. Land the cylinder cover on a couple of wooden planks on the upper platform.

9.



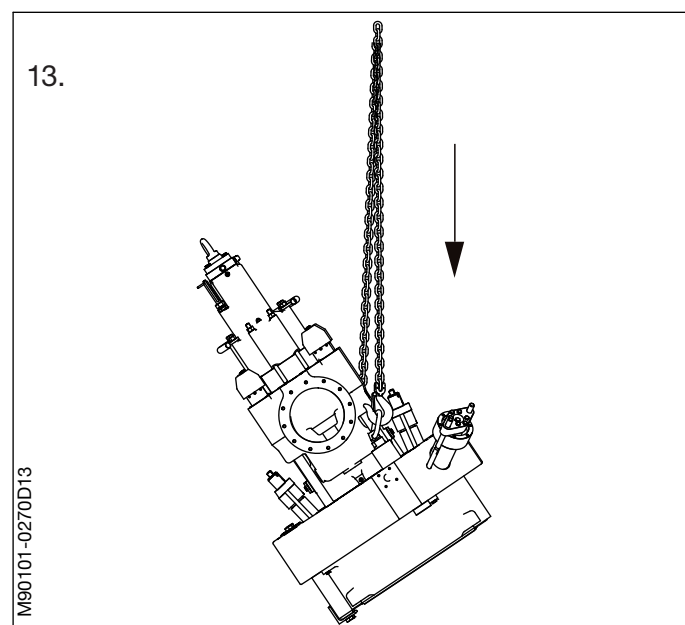
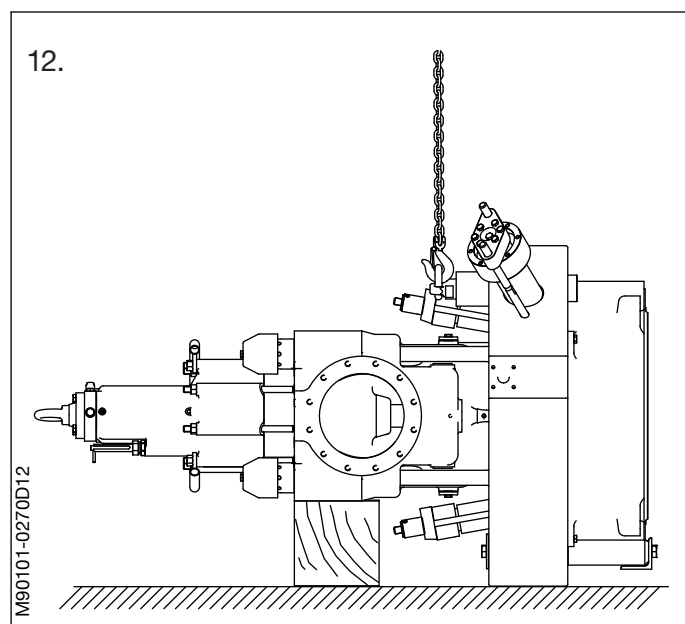
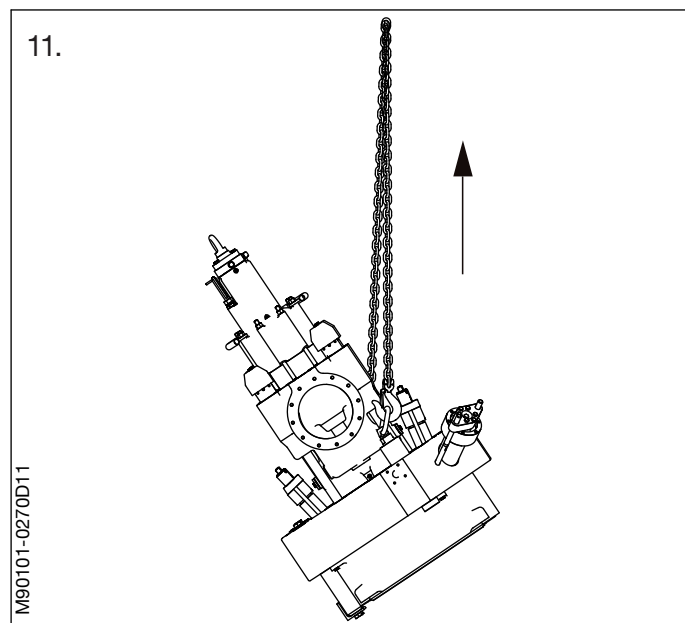
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10.

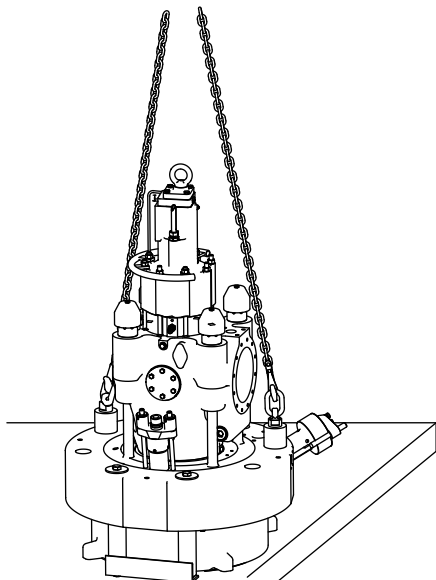


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11. Hook on to the lifting tool with a strap or a set of lifting chains, and lift the cylinder cover with the engine room crane to an angle of approx. 45 degrees.
12. Apply pressure (by hand) to the exhaust valve housing while lowering the engine room crane in order to tilt the cylinder cover into a horizontal position and place a wooden plank to support the exhaust valve housing.
13. When turning the cylinder cover to vertical position, hook on the lifting tool with a strap or a set of lifting chains and lift the cylinder cover with the engine room crane to an angle of approx. 45 degrees.



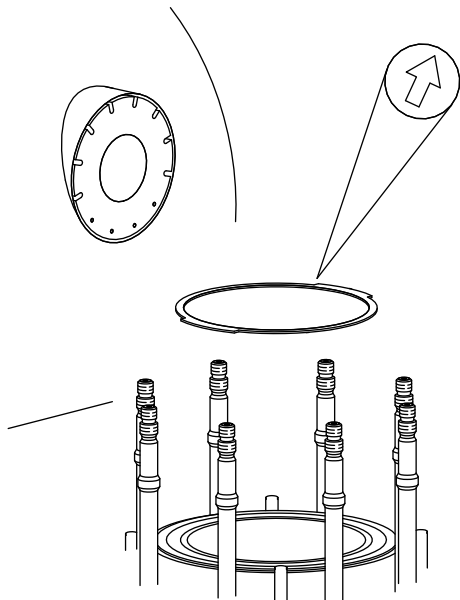
14.



14. Apply pressure (by hand) to the cylinder cover while lowering the engine room crane and follow the movement of the cylinder cover with the crane until the cylinder cover is in a vertical position.

15. Remove and discard the sealing ring between the cylinder cover and the cylinder liner.

15.



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1. Dismount all valves and pipes on the cylinder cover.

Dismount all studs on top of the cylinder cover.

2. Remove the four screws which secure the cooling jacket to the cylinder cover.

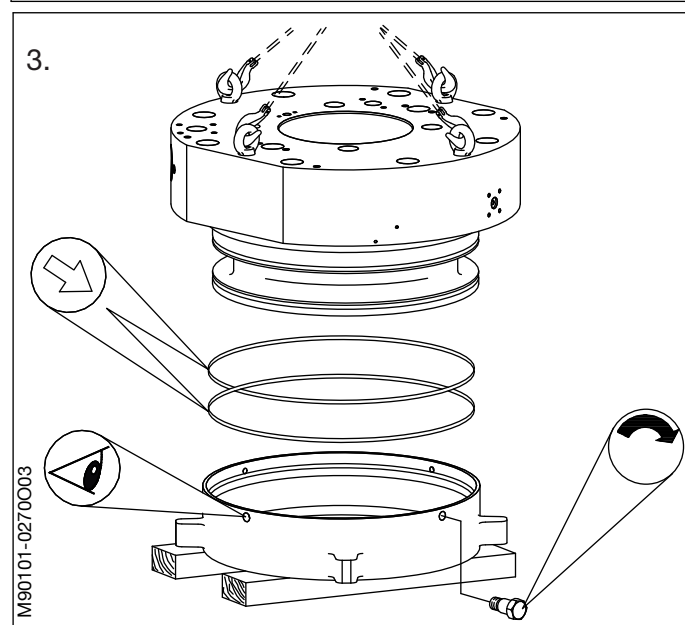
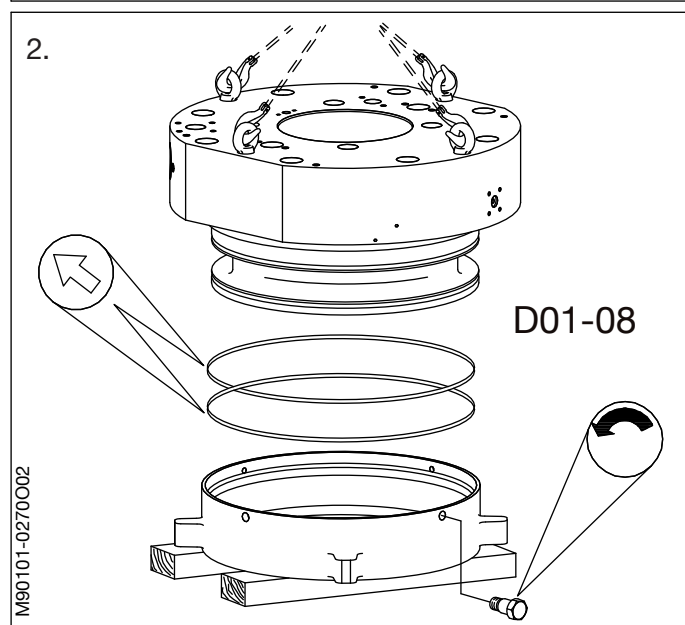
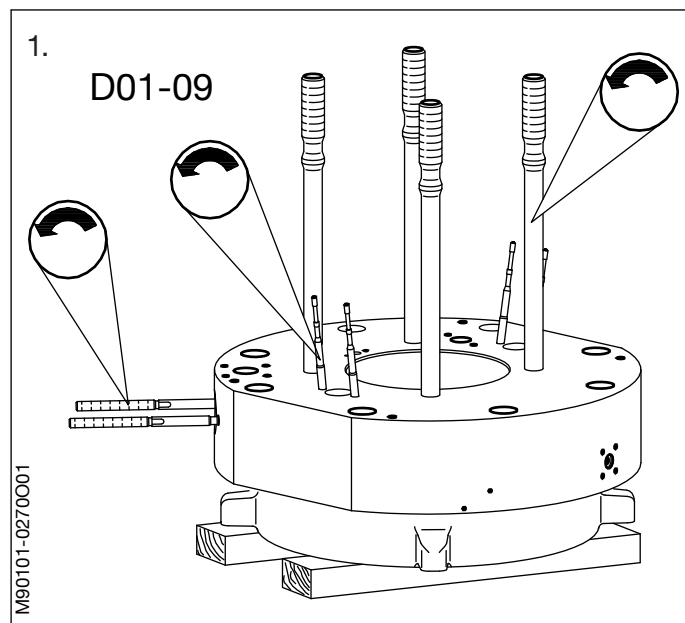
Lift the cylinder cover free of the cooling jacket and land it on a couple of planks.

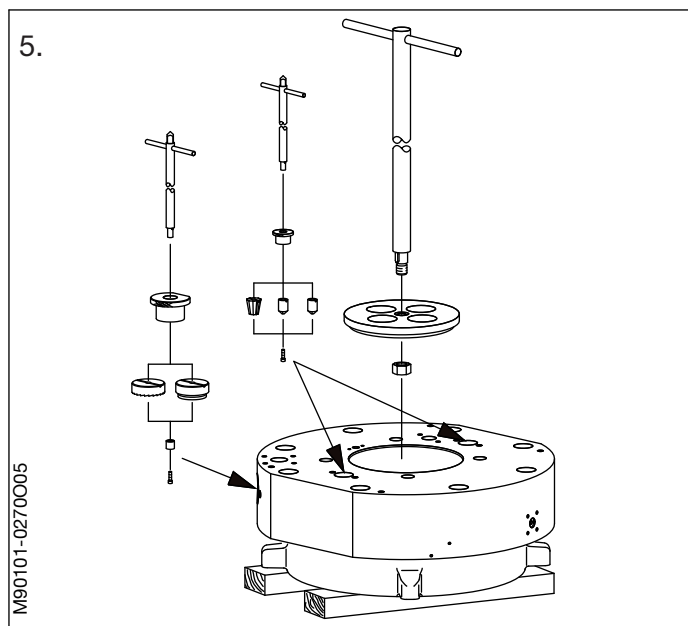
Remove and discard the O-rings from the cover, and carefully clean the cooling jacket and the cylinder cover.

3. Provide the cylinder with new O-rings, well lubricated with oil.

4. Lift the cylinder cover and guide the jacket/cover into position, using the guide pin as a reference (Camshaft side).

Mount and tighten the screws for the cooling jacket.





5. Recondition the valve bores as follows:

Fuel valve bore

6. The tool consists of a common spindle with handle, a guide, a carbon cutter, a seating face cutter and a grinding mandrel.

Clean the fuel valve bores, using the carbon cutter. If required, recondition the fuel valve seating with appropriate cutter.

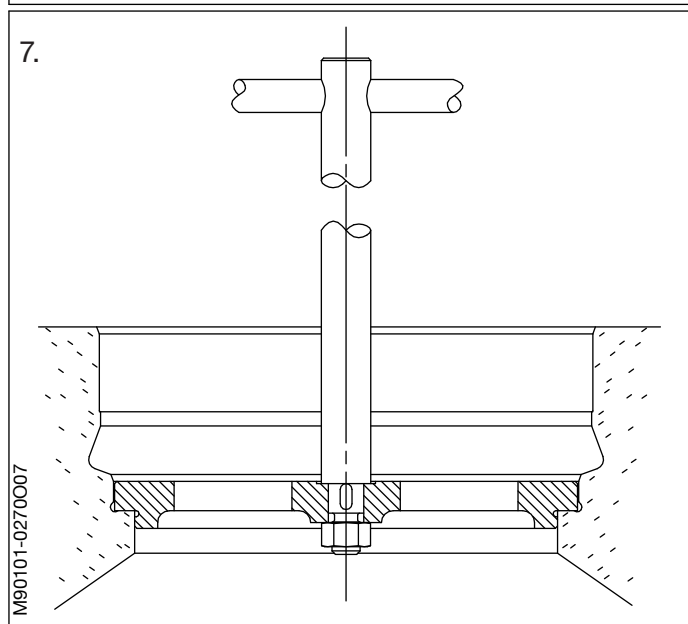
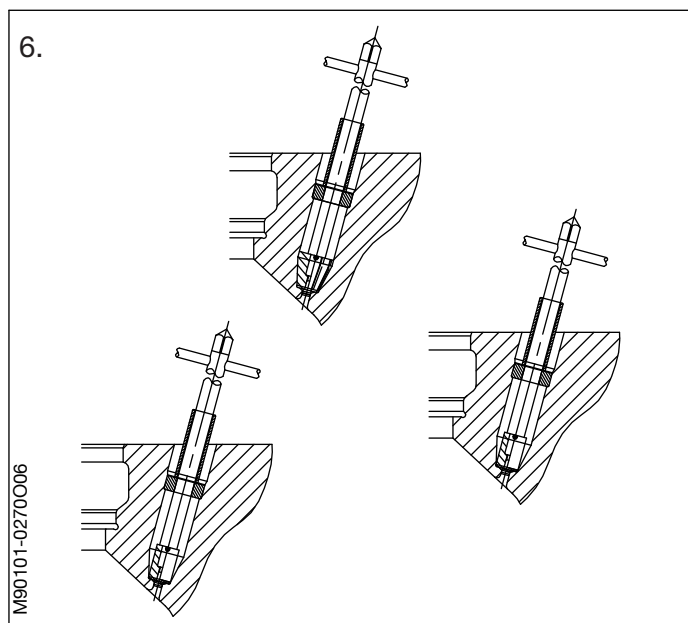
Grind the seating with a grinding mandrel and a grinding compound (e.g. Carborundum No. 200).

After the milling/grinding, clean the bore and seating carefully, and check that the seating is not damaged.

7. The tool consists of a spindle with handle a grinding disc.

After cleaning the valve boare and seating, grind the seating with a grinding mandrel and a grinding compound (e.g. Carborundum No. 200).

After the grinding, clean the bore and seating carefully, and check that the seating is not damaged.



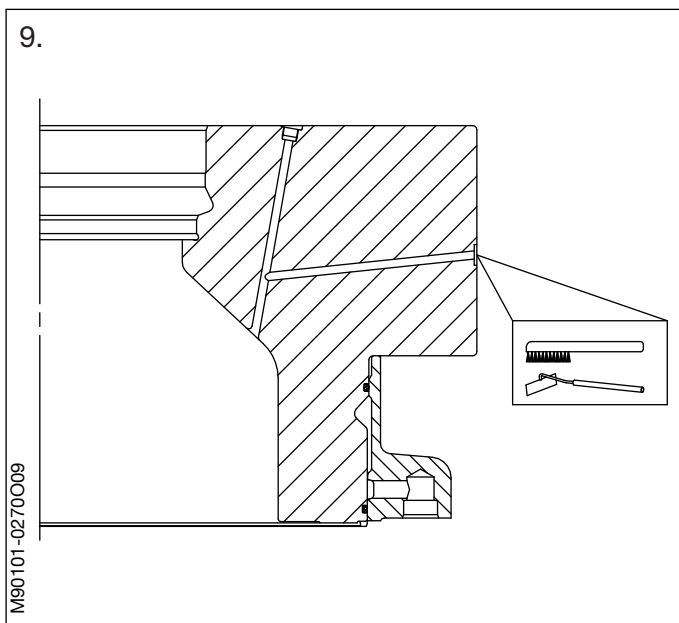
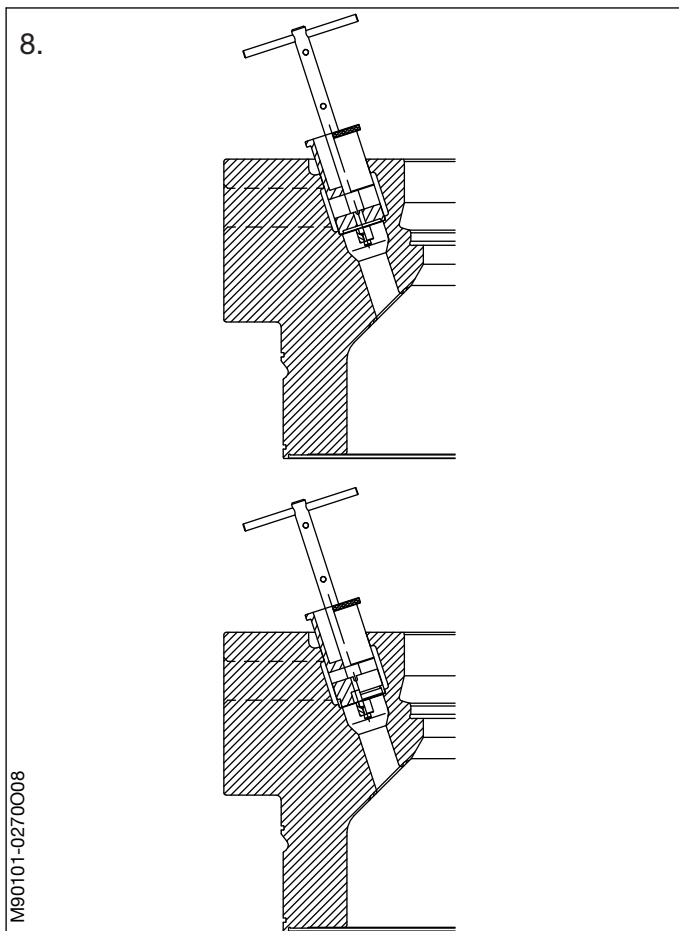
Starting valve bore

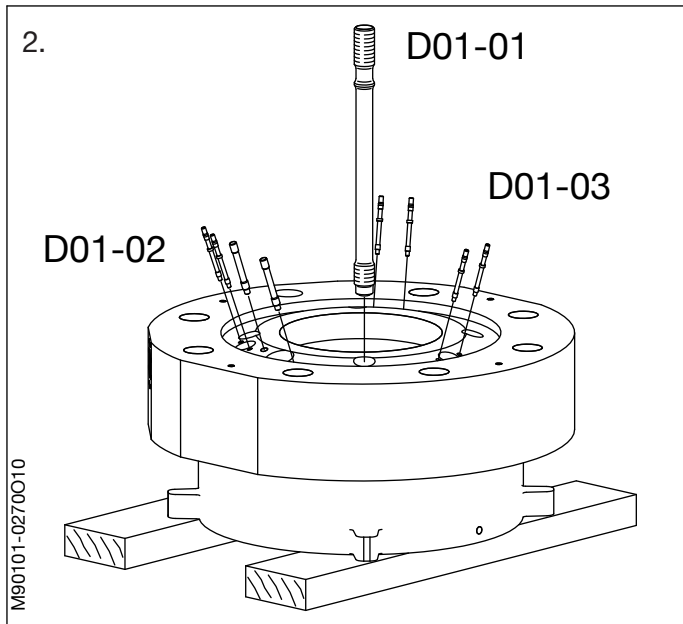
8. The tool consists of a guide, a cutter and a grinding disc.

Recondition the starting valve bore and seating in the same way as described for the fuel valve bore.

When replacing valves on a cylinder cover that is mounted on the engine, recondition the valve bores/seating in the same way as described here, but without dismantling the studs.

9. Clean the bores for safety valve/indicator cock.





10. Lubricate the threads with 'Never Seize' or Molybdenum Disulphide (MoS_2) and mount the valve studs.

Tighten up the studs in accordance with the screwing-in torque stated in Data.

Fill up the grooves between the valve studs and the bores for the valve studs with permatex to prevent water or oil from entering the bores during operation of the engine.

Provide the water connection flanges with new gaskets, and keep the cylinder cover ready for replacement.

Always fit new or overhauled valves before a cylinder cover is mounted on the engine.

1. Place a new sealing ring on top of the cylinder liner with the guide tabs in the fore and aft direction.

This is to ensure that the tabs fit into the two gas blow-out grooves milled in the cylinder cover guide recess.

2. Lift the cylinder cover by means of the crane and **carefully** clean the contact surface which faces the cylinder liner.

Provide the cooling water connection pipes with new O-rings and lubricate them with grease or soft soap.

Mount a new 'klinger' sealing ring in the groove of the flange of the intermediate pipe.

Lower the cover **carefully** into position.

During the landing, **carefully** check that the cooling water connection pipes engage correctly with the holes in the cooling jacket.

3. Mount the cover nuts on the studs and screw them down.

Place eight hydraulic jack supports, one around each nut, and mount the eight hydraulic jacks on the studs.

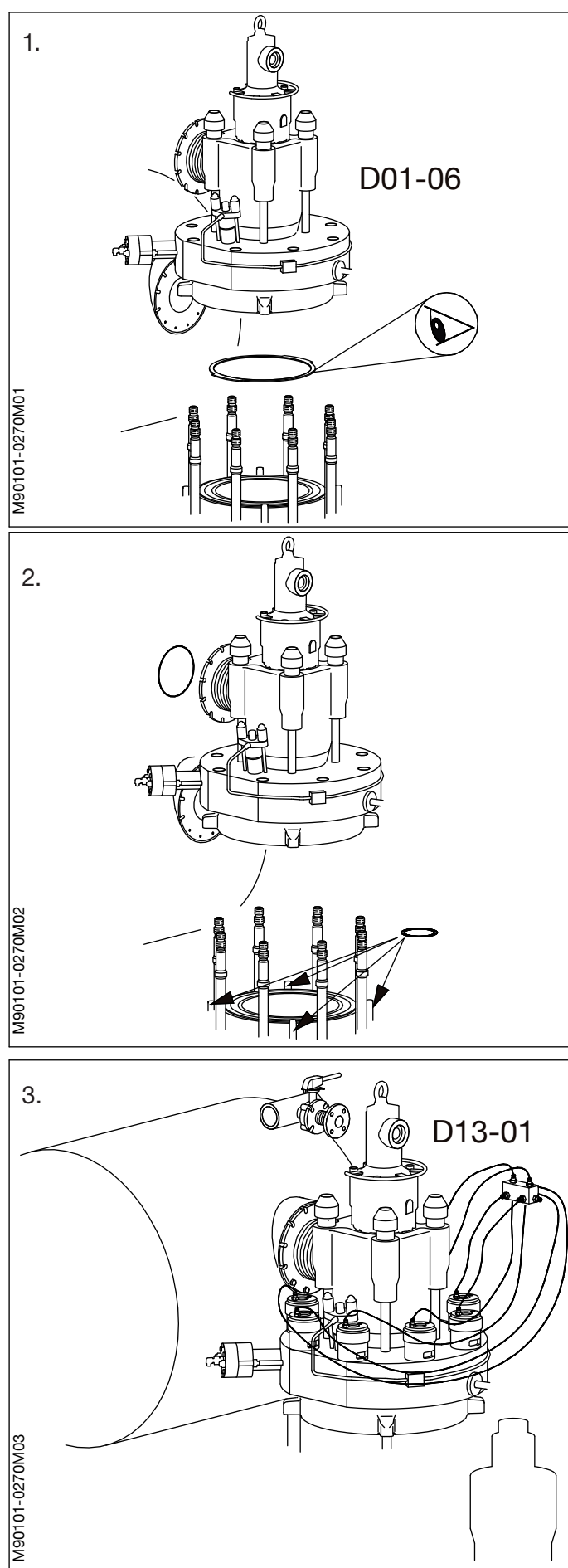
Connect the high-pressure pump to the jacks by means of the distributor block and the high-pressure hoses.

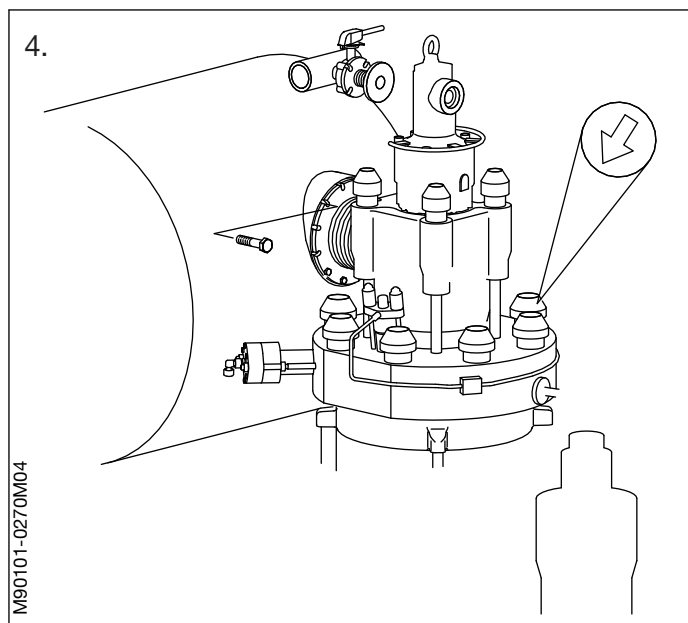
Using the vent screws of the hydraulic jacks, vent the system, and then increase the pressure as stated in Data. For operation of the hydraulic tools, see Procedure 913-1.

Tighten the nuts by applying the tommy bar through the slots in the hydraulic jack supports.

Relieve the system of pressure.

Finally, remove the hydraulic jacks and their supports from the nuts.





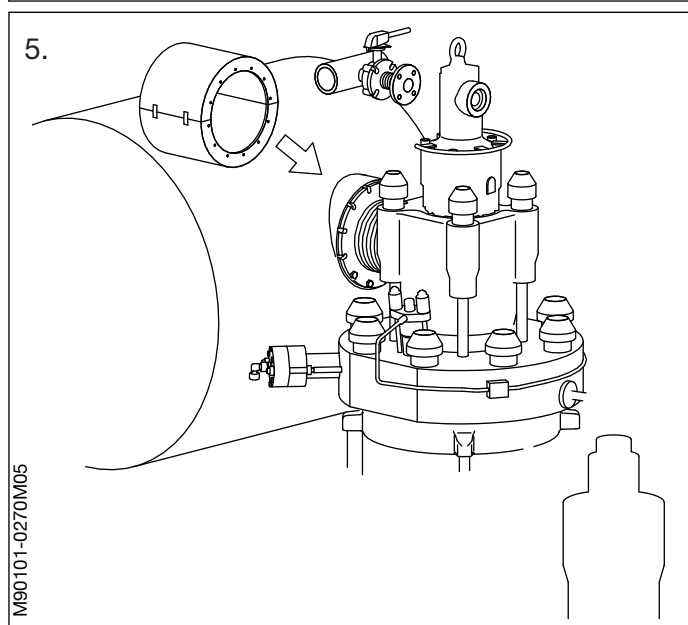
4. Mount the protective caps on the cylinder cover studs.

Align the intermediate pipe so that the holes in the flange of the intermediate pipe are in line with the threaded holes in the exhaust receiver.

Mount the screws of the flange between the intermediate pipe and the exhaust receiver.

Note!

Lubricate the screws with "Never Seize" or molybdenum disulphide (MoS_2) before fitting.



5. Mount the insulating jacket around the intermediate pipe.

6. Mount the return oil pipe and the air pipe for pneumatic closing of the exhaust valve.

Mount the high-pressure pipe for the hydraulic valve gear.
See Procedure 908-1.4.

7. Mount the starting air pipe and control air pipe for the starting valve.

Mount the cooling water inlet and outlet pipes on the exhaust valve.

Mount the drain pipe on the intermediate pipe between the exhaust valve and the cylinder cover.

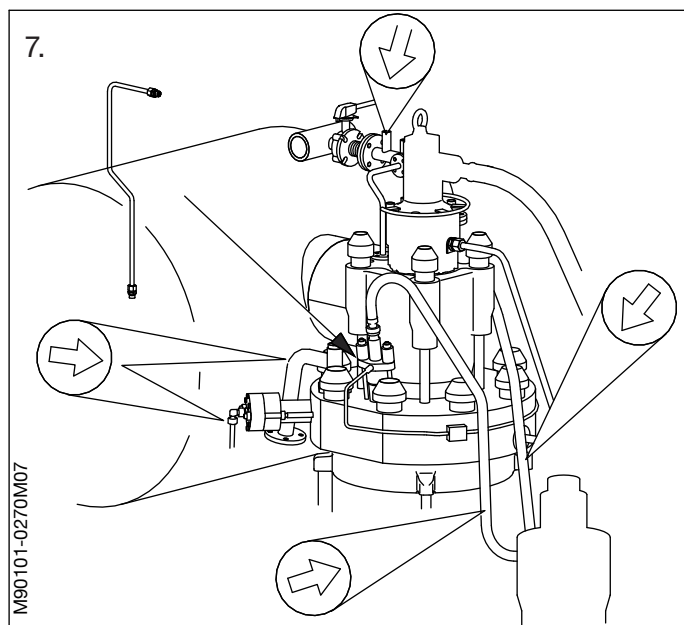
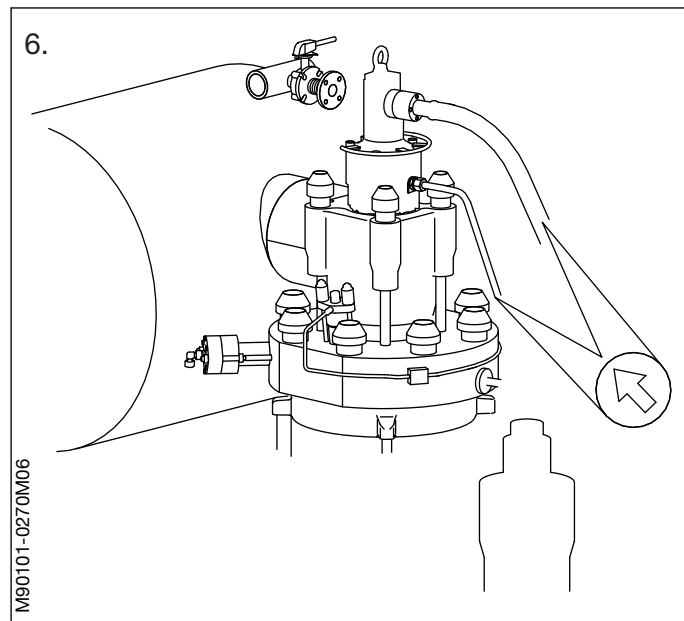
Mount the fuel oil high-pressure pipes for the fuel valves.

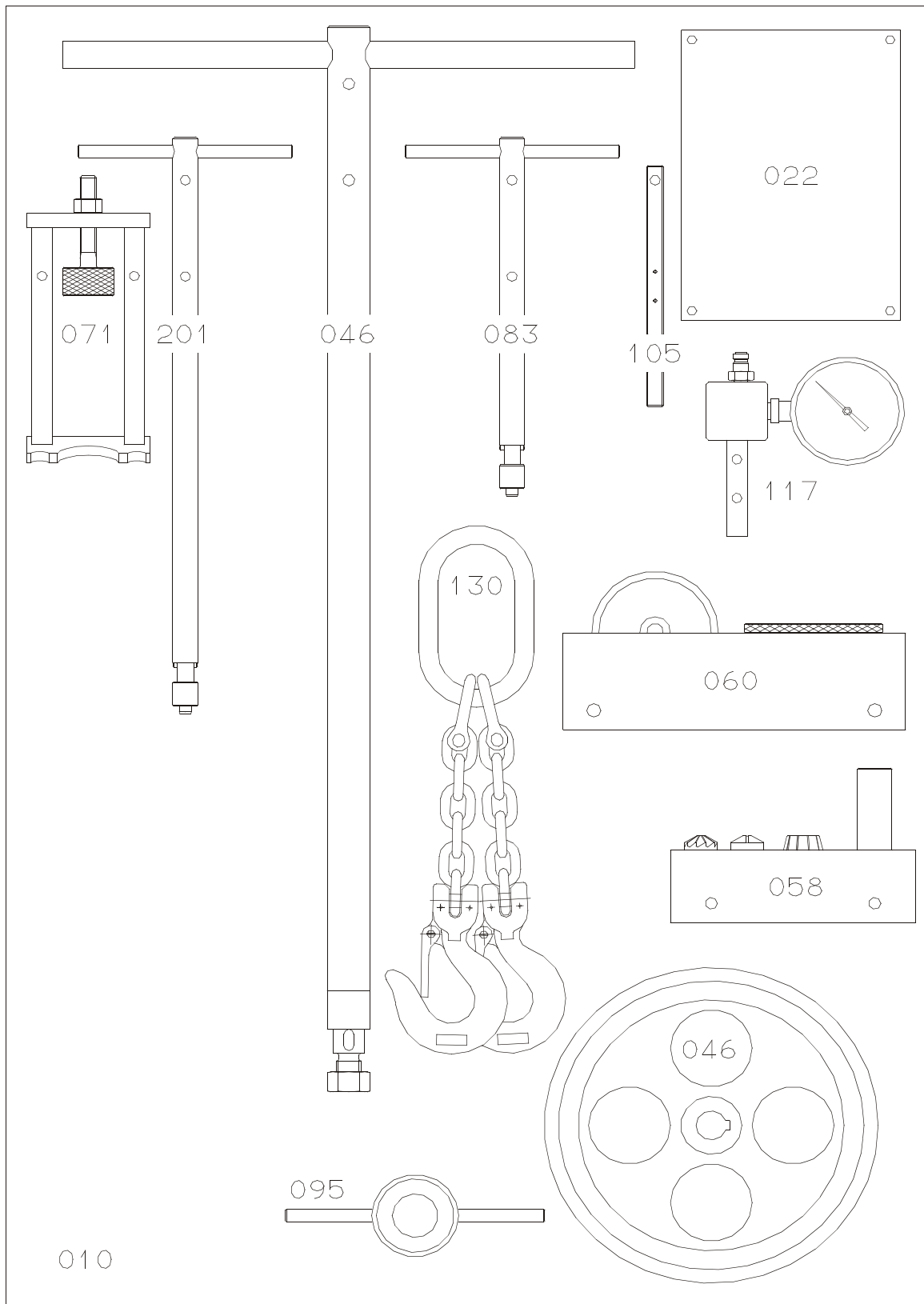
See Procedure 909-14.4.

Connect the outlet pipe from the return oil pipe on the fuel valves.

Shut the drain valves and open the cooling water inlet. After venting the cylinder section, shut the vent cocks and open the cooling water outlet valve.

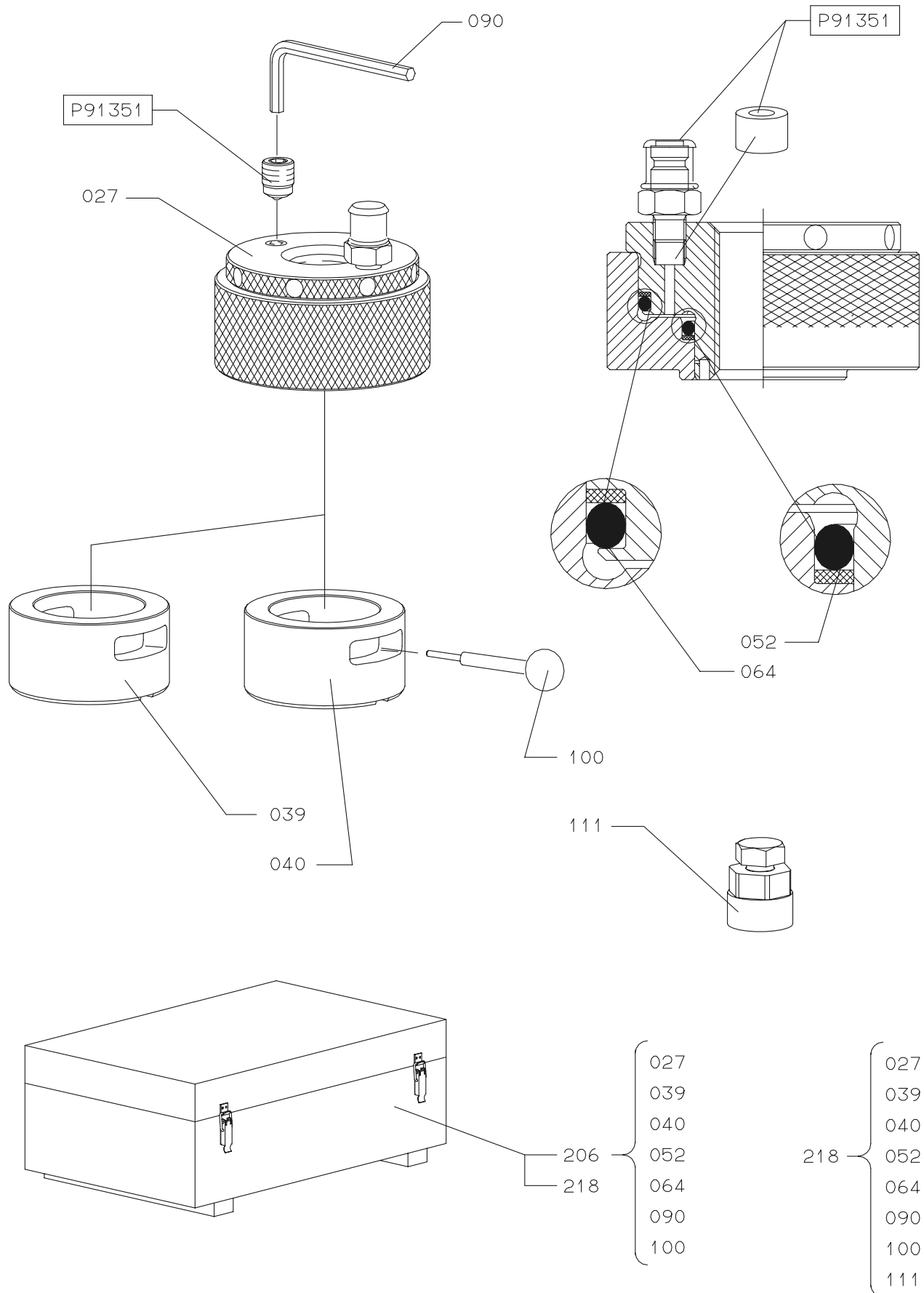
Open the fuel oil, lubricating oil and air supplies to the alarm system.





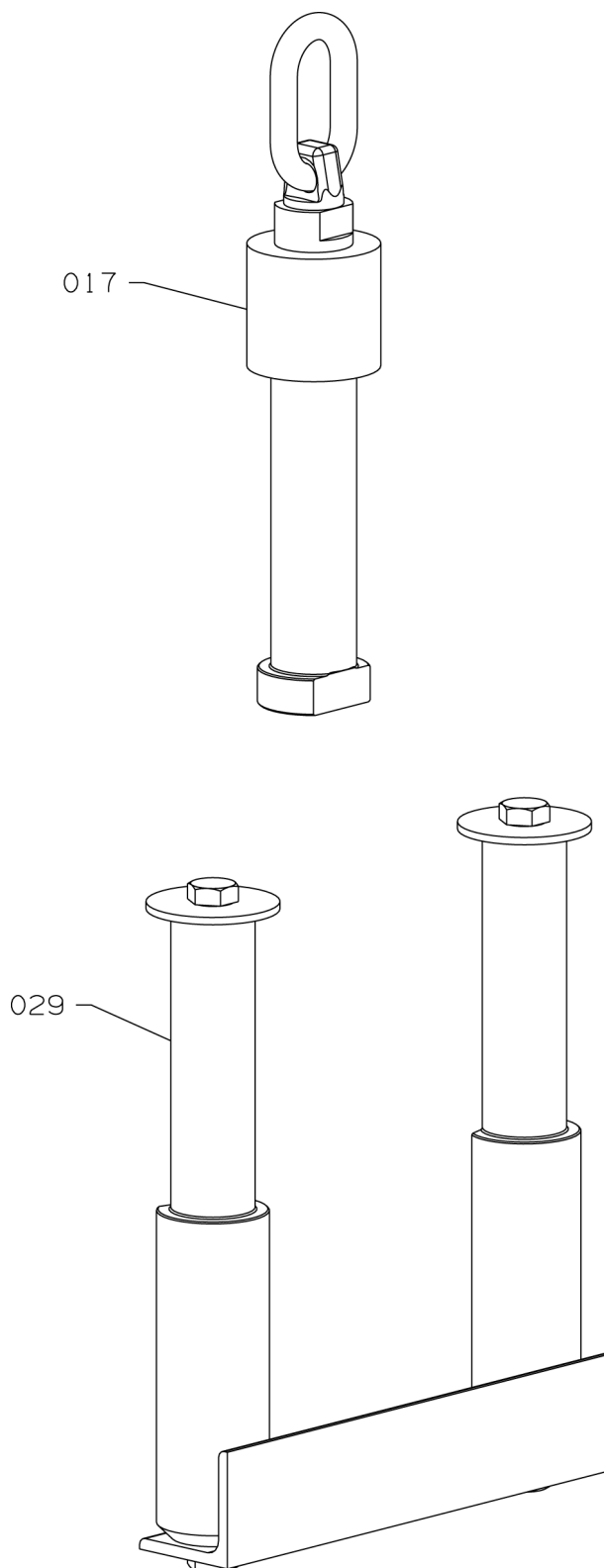
Item No.	Item Description
010	Panel for tools
022	Name plate
046	Grinding tool for exhaust valve seat
058	Milling and grinding wheel for fuel valve seat
060	Milling and grinding wheel for starting valve seat
071	Dismantling tool for fuel valve seat
083	Handle for cutting and grinding tool
095	Grinding ring for starting valve
105	Grinding handle
117	Pressure tester for safety valve
130	Lifting chains
201	Handle for cutting and grinding tool

Item No.	Item Description





Item No.	Item Description	Item No.	Item Description
027	Hydraulic jack, complete		
039	Support		
040	Support		
052	O-ring with back-up ring		
064	O-ring with back-up ring		
090	Spanner		
100	Tommy bar		
111	Stud setter		
206	Hydraulic tool set, complete		
218	Hydraulic tool set, complete		



Item No.	Item Description	Item No.	Item Description
017 029	Lifting tool for cylinder cover Lifting tool for cylinder cover		